

BóSight: Multimodal Deep Learning for Individual Dairy Cow Behaviour and Health-Risk Screening

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Abstract. Early signs of illness, lameness, and metabolic stress in dairy cows are subtle and gradual, and continuous manual observation of every animal in a herd is impractical at commercial scale. Overhead cameras can capture what a cow *does*, while wearable sensors capture its internal physiological state, but neither modality alone tells the full story. We present BóSight, an end-to-end seven-stage pipeline that fuses computer vision and wearable-sensor data to screen individual dairy cows for health risk automatically. The pipeline detects every cow in an overhead frame (YOLOv8s, $\text{mAP}_{50} = 0.992$), re-identifies each animal from its coat pattern (ResNet-50, 97.0% accuracy), tracks identities across a full day using an appearance-based approach that requires no motion model (98.4% end-to-end accuracy), classifies behaviour into four daily activities (ResNet-50, 95.0% accuracy, macro-F1 0.807), estimates body condition from a geometric proxy, fuses these vision outputs with inertial, temperature, and positioning sensor streams, and screens each cow against seven veterinary-literature thresholds to produce an interpretable, traceable health alert. Evaluated on the MmCows dataset, the pipeline is demonstrated through case studies in which three at-risk cows were each identified through a *different* signal — behaviour, physiology, and body condition respectively — providing direct evidence that multimodal fusion detects risks that no single modality would catch alone. We report the pipeline’s quantitative performance, discuss the honest limitations imposed by a single-day, single-camera dataset with no clinical ground truth for body condition or health events, and outline concrete directions for developing the system toward deployable, on-farm precision livestock monitoring.

Keywords: Precision Livestock Farming · Computer Vision · Multimodal Sensor Fusion · Behaviour Classification · Animal Re-identification · Dairy Cattle Health Monitoring

1 Introduction

Modern dairy farming operates at a scale where continuous manual observation of every animal is impractical. Yet many health problems — lameness, metabolic stress, illness, and reduced feeding — first manifest as subtle, gradual changes

in an individual cow’s behaviour and physiology, well before they are visible to a human observer. By the time a cow is visibly unwell, animal welfare and milk yield have often already suffered. Precision Livestock Farming (PLF) aims to close this gap through continuous automated sensing.

The central challenge is that no single data source tells the complete story. Overhead cameras observe what a cow *does* — its posture, its location, its interactions with feed and water — but reveal nothing of its internal physiological state. Wearable sensors measure movement and core body temperature directly, but provide no behavioural context and, in practice, are only economically deployed on a subset of a herd. A monitoring system built on either modality alone therefore has a structural blind spot.

We present **BóSight**¹, a seven-stage pipeline that fuses overhead-camera computer vision with wearable inertial, temperature, and positioning sensors to screen individual dairy cows for health risk. The pipeline is evaluated end-to-end on the MmCows dataset [1], a multimodal dairy-cattle deployment providing camera footage, one fully hand-annotated day of behaviour, and wearable-sensor streams for a subset of the herd. Holstein-Friesian, the breed in MmCows, is also Ireland’s dominant dairy breed, making the setting directly relevant to Irish herds and the wider VistaMilk research programme.

Our contributions are:

1. An end-to-end pipeline spanning detection, individual re-identification, appearance-based tracking, behaviour classification, body-condition estimation, multi-modal sensor fusion, and interpretable rule-based health-risk screening, presented in an interactive dashboard.
2. Empirical evidence, via individual case studies, that fusing vision and wearable-sensor signals surfaces at-risk animals that would be missed by either modality alone.
3. An explicit, honest treatment of what a single-day, single-camera dataset with no clinical ground truth for body condition or health events does and does not permit, and a concrete accounting of what additional data would be required to move from proof-of-concept to a validated, deployable system.

2 Related Work

Camera-based cattle monitoring. Single-camera vision systems have been applied successfully to specific cattle health tasks; Khan et al. [2] developed a real-time lameness-detection system from a single side-view camera, demonstrating that a single overhead or side-on view can carry clinically meaningful signal without wearable instrumentation.

Individual animal re-identification. Distinguishing individual cattle by coat pattern using deep metric learning was established by Andrew et al. [3] for Holstein-Friesian cattle, showing that natural coat markings are a sufficiently discriminative, non-invasive identity signal — motivating BóSight’s use of

¹ *Bó* is the Irish word for cow.

appearance-based re-identification in place of physical tagging or motion-based tracking.

Automated behaviour recognition. Yu et al. [4] applied deep learning to automatic recognition of daily dairy cow behaviours, demonstrating the feasibility of vision-based behavioural ethograms at a granularity comparable to the four-class scheme used in this work.

Multimodal sensor fusion for health. Most closely related to our contribution, Paulauskaite-Taraseviciene et al. [5] investigated AI-driven multimodal sensing for the early detection of health disorders in dairy cows, supporting the central premise of this paper: that combining sensing modalities improves early detection of health disorders beyond what any single modality achieves.

BóSight differs from this body of work in integrating detection, re-identification, tracking, behaviour classification, body-condition estimation, and multimodal sensor fusion into a single, interpretable, end-to-end pipeline evaluated on one public dataset, and in explicitly demonstrating — through paired case studies — *which* modality catches which type of risk.

3 Dataset

BóSight is evaluated on **MmCows** [1], a multimodal dataset comprising a deployment of 16 Holstein-Friesian dairy cows. The dataset provides:

- **Vision:** overhead footage from camera `cam_1`, approximately 5,040 frames over 24 hours, with one fully hand-annotated day (25 July 2023) providing ground-truth bounding boxes, individual cow identities, and per-second behaviour labels.
- **Wearable sensors** (10 of the 16 cows only): inertial measurement unit (IMU) acceleration, core body temperature (CBT), and ultra-wideband (UWB) indoor positioning.

The remaining six cows (C11–C16) have no wearable instrumentation and are vision-only throughout the pipeline; this partial sensor coverage is treated explicitly at every downstream stage rather than papered over.

Three properties of the dataset directly shape the design decisions in Section 4: (i) frames are captured roughly 17 seconds apart, too sparse for conventional motion-prediction tracking; (ii) behaviour classes are severely imbalanced, with the rarest class (locomotion) comprising approximately 1.5% of annotated samples; and (iii) the dataset provides no clinical ground truth for body condition score or for health events within the deployment window, constraining Stages 5 and 7 to interpretable proxies rather than trained, validated models (Section 7).

Milk yield records exist in MmCows but were deliberately excluded from the fused feature set: with only a single annotated day, there is no per-animal historical baseline against which a yield *trend* — the clinically meaningful signal — can be computed, and cross-animal comparison on a single day confounds genuine health signal with natural variation in age and lactation stage.

4 Methodology and System Architecture

BóSight is structured as seven sequential stages, each consuming the output of the one before it, moving from a raw overhead frame to a traceable, per-cow health alert (summarised, with results, in Table 2).

4.1 Stage 1 — Detection

A YOLOv8s object detector was fine-tuned on a single "cow" class using the dataset's ground-truth bounding-box annotations. Given a raw overhead frame, the detector performs a single forward pass and returns a set of candidate bounding boxes with confidence scores; overlapping duplicate boxes on the same animal are resolved via non-maximum suppression. Frames in this deployment typically contain 8–13 visible cows simultaneously.

4.2 Stage 2 — Re-identification

Each detected bounding box is cropped from the source frame and passed to a ResNet-50 classifier fine-tuned (from ImageNet-pretrained weights) as a 16-way classification head over the individual cow identities C01–C16, trained on 53,508 ground-truth crops cut from the dataset's human-annotated boxes. The model outputs a single predicted identity per crop.

4.3 Stage 3 — Tracking

Conventional motion-prediction trackers (e.g. ByteTrack, BoT-SORT) assume small inter-frame displacement and are unsuitable here: at a ~ 17 -second frame interval, cows move too far between captures for motion prediction to be meaningful. BóSight instead relies entirely on Stage 2's appearance-based re-identification, applied independently to every frame across the day; because re-identification alone is highly accurate, per-frame identity assignment provides the temporal continuity that a motion model would otherwise supply, with no explicit tracking algorithm required.

4.4 Stage 4 — Behaviour Classification

A second ResNet-50, again fine-tuned from ImageNet-pretrained weights, was trained as a four-way classifier over lying, standing, feeding, and moving, using the same crop images as Stage 2 but labelled by joining each crop's embedded timestamp against the dataset's per-second behaviour annotations rather than by folder identity. The severe class imbalance noted in Section 3 was addressed with a weighted random sampler (oversampling rare classes during training) combined with class-weighted cross-entropy loss. The model was trained for 15 epochs (Adam, learning rate 10^{-4} , batch size 64, with `ReduceLROnPlateau` scheduling), and the checkpoint with the best validation macro-F1 was retained to avoid selecting an over-fit final-epoch model.

Predictions for a given cow across the full day are aggregated into a daily behaviour time-budget — the fraction of observed time spent in each of the four classes — which forms one of the inputs to Stage 6.

4.5 Stage 5 — Body Condition Estimation

MmCows provides no body-condition-score (BCS) ground truth. Rather than omit body condition entirely, BóSight computes a transparent geometric proxy: for each cow, the median bounding-box area across its *standing*-posture detections (18,124 detections across the herd) is min–max scaled across the herd into a conservative 2.0–4.0 band, deliberately avoiding the extremes of the clinical 1–5 BCS scale that a proxy cannot justify. This yields a *relative* within-herd ranking rather than a calibrated clinical score, a limitation discussed further in Section 7.

4.6 Stage 6 — Multimodal Sensor Fusion

For each cow-day, Stage 6 constructs a single fused feature vector combining:

- **Vision:** the Stage 4 behaviour time-budget and the Stage 5 body-condition estimate (all 16 cows).
- **IMU:** mean and standard deviation of tri-axial acceleration magnitude, and an *active fraction* — the proportion of samples whose acceleration magnitude deviates by more than 0.5 m/s^2 from the stationary gravity baseline of 9.81 m/s^2 (C01–C10 only).
- **CBT:** daily mean, maximum, minimum, and standard deviation, computed after excluding physiologically implausible readings ($< 35^\circ\text{C}$ or $> 42^\circ\text{C}$) attributable to sensor dropout (C01–C10 only).
- **UWB:** total distance travelled and spatial range, derived from consecutive-position displacement (C01–C10 only).

Cows without wearable instrumentation (C11–C16) receive vision-only feature vectors, explicitly flagged rather than imputed, ensuring downstream stages treat missing sensor coverage honestly rather than silently.

4.7 Stage 7 — Health-Risk Screening

No health-event labels exist within the MmCows deployment window (the dataset’s health records are lifetime histories spanning 2014–2024, with no medically relevant event near the annotated day), precluding supervised training of a health classifier. Stage 7 is therefore implemented as a transparent, veterinary-knowledge rule-based screening system: each cow accumulates a severity score from seven threshold-based flags (Table 1), each grounded in veterinary-literature reference ranges. A cow with severity ≥ 3 is flagged *critical*, 1–2 *suspect*, and 0 *healthy*; every flag is traceable to the specific rule and sensor value that triggered it.

Table 1. Health-risk screening rules: weighted severity scoring against veterinary-literature thresholds. Severity ≥ 3 : critical; 1–2: suspect; 0: healthy. Sensor-dependent rules are inactive for vision-only cows.

Flag	Threshold	Severity
Fever	CBT max $\geq 39.5^\circ\text{C}$	3
Reduced feeding	$\leq 15\%$ of day	2
Lethargy	activity $z \leq -1.5$	2
Elevated temperature	CBT mean $\geq 39.0^\circ\text{C}$	1
Restlessness / estrus	activity $z \geq 2.0$	1
Excessive lying	$\geq 60\%$ of day	1
Low body condition	BCS proxy ≤ 2.2	1

4.8 Dashboard

Stages 1–7’s precomputed outputs are surfaced through an interactive Streamlit dashboard performing no live model inference. A *Herd Overview* page presents alert-level summaries, the herd-wide behaviour distribution, and a sortable, colour-coded table across all 16 cows; a *Per-Cow Detail* page presents a reference photo, the 24-hour behaviour distribution, sensor readings, and a rule-by-rule breakdown of exactly which of the seven Stage 7 flags fired and why (Fig. 2 and Fig. 3).

5 Experimental Results

5.1 Detection, Re-identification, and Tracking

Table 2 reports test-set performance for the three vision stages with available ground truth. The detector achieves $\text{mAP}_{50} = 0.992$; re-identification achieves 97.0% test accuracy over the 16-way identity classification task; and the appearance-based tracking approach — despite using no motion model — achieves 98.4% end-to-end accuracy (99.29% detection rate combined with 99.07% identity accuracy on matched detections), directly supporting the design choice discussed in Section 4 to substitute high-accuracy re-identification for motion-based association under sparse temporal sampling.

5.2 Behaviour Classification

The behaviour classifier achieves 95.0% overall accuracy and macro-F1 0.807 (weighted-F1 0.951) on the held-out test split. Per-class F1 scores are: *lying* 0.997, *feeding* 0.959, *standing* 0.933, and *moving* 0.339 (Fig. 1). Three of the four classes exceed 0.93 F1; *moving* is the clear outlier, and its confusion is concentrated with *standing*. We attribute this specifically to locomotion being a *temporal* behaviour that a single still frame cannot reliably capture — a walking cow and a standing cow can be visually near-identical in one frame — rather than

Table 2. Quantitative results by pipeline stage on the MmCows held-out test split. Body condition and health-risk alerts have no ground truth in MmCows and are reported qualitatively rather than as validated metrics (Section 7).

Stage	Method	Key result	Ground truth?
Detection	Fine-tuned YOLOv8s	mAP ₅₀ = 0.992	Yes
Re-identification	ResNet-50, 16-class	97.0% accuracy	Yes
Tracking	Appearance-based	98.4% overall accuracy	Yes
Behaviour classification	ResNet-50, 4-class	95.0% acc. / macro-F1 0.807	Yes
Body condition	Geometric proxy	Relative ranking (2.0–4.0)	No
Health-risk alerts	Rule-based, 7 flags	12 / 3 / 1 (healthy/suspect/critical)	No

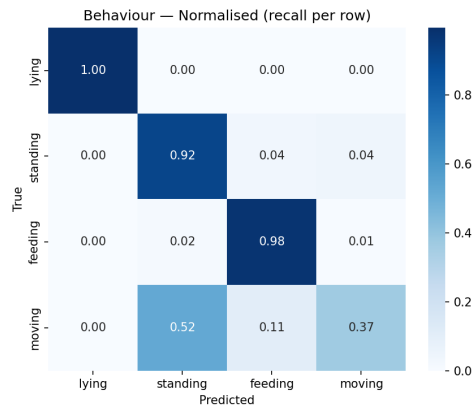


Fig. 1. Behaviour classifier confusion matrix on the held-out test split, row-normalised by true class (recall per class). Three of four classes exceed 0.93 F1; *moving* is the outlier.

to a deficiency of the class-imbalance handling; this is corroborated by the fact that the confusion is asymmetric toward *standing* specifically, the behaviourally closest class, rather than distributed uniformly. Section 7 discusses temporal modelling and IMU fusion as the direct remedy.

5.3 Health-Risk Screening Output

Applying Stage 7 to the fused per-cow feature vectors (Section 4.7) over the 16-cow herd yields 12 cows flagged *healthy*, 3 *suspect*, and 1 *critical*.

5.4 Case Studies

To evaluate whether multimodal fusion provides value beyond any single modality, we trace four cows individually end-to-end through the full pipeline.

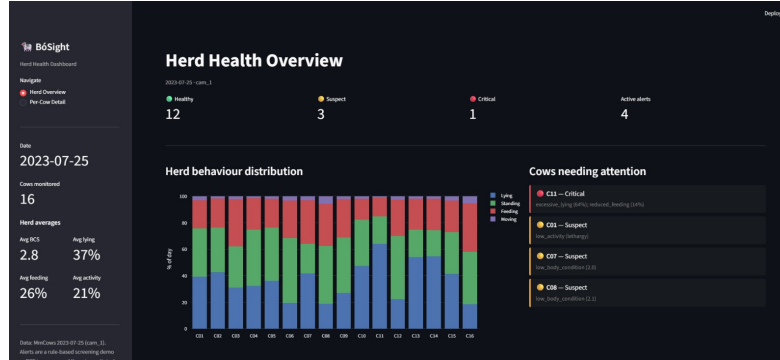


Fig. 2. Dashboard Herd Overview: alert-level summary and colour-coded per-cow status across the full 16-cow herd.

C11 (critical, behavioural). A vision-only cow (no wearable instrumentation). Excessive lying (64% of the day) and reduced feeding (14% of the day) each trigger a rule, yielding severity 3. Because C11 has no wearable sensors, *only the camera-derived behaviour signal* was able to flag this animal.

C01 (suspect, physiological). Behaviour is within normal range for this cow, but its wearable IMU records the lowest activity fraction (15%) in the herd, triggering the lethargy rule (severity 2). Here, *only the wearable sensor* caught the anomaly — the camera-derived behaviour signal alone would have missed it.

C07 (suspect, body condition). Feeding and movement are both within normal range, but the body-condition proxy is the herd’s lowest (2.00), triggering the low-body-condition rule (severity 1).

C05 (healthy baseline). A behaviourally balanced day, body-condition proxy 3.65, and all available sensor readings within the herd’s normal range; no rules fire.

These four cows were flagged, or correctly left unflagged, through three *different* signal types — behaviour, physiology, and body condition respectively — providing direct, concrete evidence that fusing modalities detects risk that no single modality would catch alone, which is the central empirical claim of this paper.

6 Discussion

The behaviour-classification results (Section 5) and the case studies together support a coherent narrative: vision and wearable sensing are *complementary*, not redundant. Vision reliably distinguishes posture-driven behaviours (lying, feeding, standing) but is structurally weak at capturing locomotion from a single frame; wearable IMU sensing measures motion directly and is precisely the modality that would resolve this weakness, but is available for only 10 of 16 cows and provides no behavioural or spatial context on its own. The appearance-based



Rule	Fired	Severity if fired
Fever — CBT max $\geq 39.5^{\circ}\text{C}$	no	3
Elevated temp — CBT mean $\geq 39.0^{\circ}\text{C}$	no	1
Lethargy — activity $z \leq -1.5$	yes	2
Restless/letrous — activity $z \geq 2.0$	no	1
Excessive lying — $\geq 60\%$	no	1
Reduced feeding — $\leq 15\%$	no	2
Low body condition — BCS ≤ 2.2	no	1

Total severity score: 2 = Suspect (>3 critical >1 suspect, 0 healthy). Sensor-based rules are inactive for vision-only cows.

Fig. 3. Dashboard Per-Cow Detail: rule-by-rule breakdown showing exactly which of the seven Stage 7 flags fired for a given cow, and why.

tracking result (98.4% accuracy with no motion model) further illustrates a case where a modality substitution — replacing motion prediction with high-accuracy re-identification — was necessary given the dataset’s sparse temporal sampling, rather than a design preference.

The rule-based design of Stage 7 (Section 4.7) is a direct consequence of the absence of health-event ground truth in MmCows, not a methodological default; every flag remains traceable to a specific veterinary threshold and sensor value, which we consider a genuine advantage for a screening tool intended to support, rather than replace, a farmer’s or veterinarian’s judgement, independent of the ground-truth limitation that necessitated it.

7 Limitations and Future Work

Limitations. (i) Evaluation uses a single camera (`cam_1`) and a single annotated day, precluding assessment of multi-day behavioural trends and leaving detection subject to occlusion from the one available viewpoint. (ii) The body-condition estimate (Stage 5) is a geometric proxy, not a calibrated clinical score, and is confounded by the camera’s fixed viewing perspective. (iii) Health-risk alerts (Stage 7) are rule-based and clinically unvalidated: with no health-event ground truth available, no precision, recall, or ROC-AUC can be reported for this stage. (iv) Only 10 of 16 cows carry wearable instrumentation, so the fully multimodal feature vector is available for a subset of the herd only.

Future work. We identify three concrete directions for developing BóSight from proof-of-concept toward a deployable system. First, *real-time, on-farm deployment*: extending the offline pipeline to near-real-time inference on live multi-camera feeds, with robust long-horizon identity maintenance as animal appearance changes over time. Second, *depth-based body measurement*: replacing the Stage 5 geometric proxy with 3D/depth sensing calibrated against real weigh-scale data, yielding an angle-invariant, quantitative body-condition and growth signal. Third, *temporal behaviour modelling*: moving Stage 4 from single-frame classification to short-clip, temporal models fused with the IMU signal, directly addressing the *moving-class* weakness identified in Section 5 and expanding the

behavioural ethogram beyond the current four classes. Longer, multi-day deployments with recorded health events would additionally enable (a) reintroducing milk-yield trend as a fusion feature and (b) replacing the rule-based Stage 7 with a supervised, quantitatively validated classifier, retaining the current rule-based system as an interpretable baseline.

8 Conclusion

We presented BóSight, a complete seven-stage pipeline that moves from raw overhead video to individual, traceable, per-cow health-risk alerts by fusing computer vision with wearable-sensor data. The vision stages achieve strong quantitative performance (0.992 detection mAP₅₀, 97.0% re-identification accuracy, 98.4% end-to-end tracking accuracy with no motion model, and 95.0% behaviour-classification accuracy), while body-condition estimation and health-risk screening are implemented as transparent, interpretable proxies given the dataset’s absence of clinical ground truth for either. Through paired case studies, we demonstrated concretely that the three flagged at-risk cows in our evaluation herd were each identified through a *different* signal — behaviour, physiology, and body condition — providing direct evidence for the paper’s central claim: that multimodal fusion of vision and wearable sensing detects health risk that neither modality would catch alone.

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